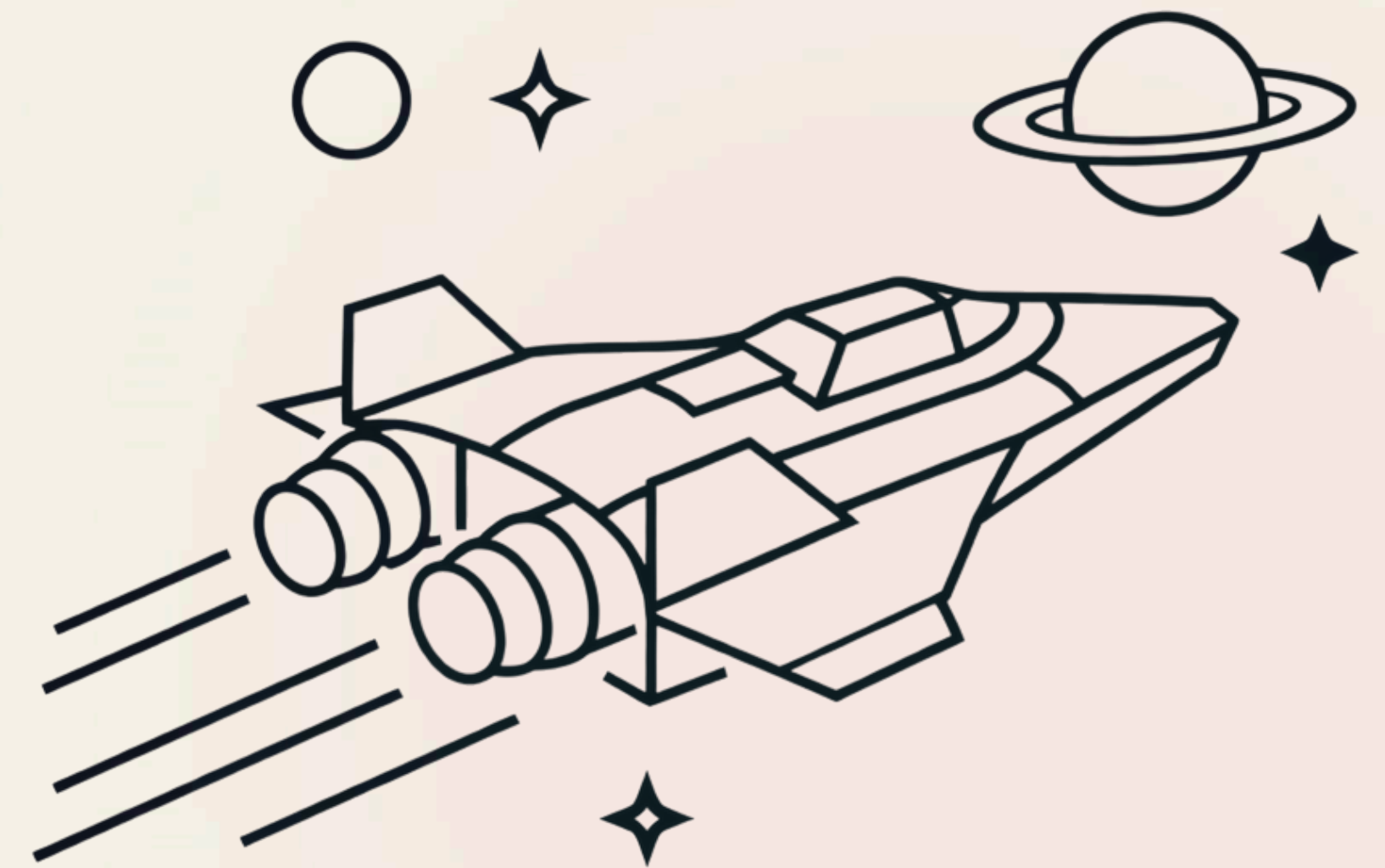


SPACESHIP X26: PASSENGER RESOURCE MANAGEMENT SYSTEM

Earth → Mars | Settlement Mission



Mission Overview:

Spaceship X26 is on a mission from Earth to Mars, carrying passengers who plan to settle on Mars. During the journey, passengers rely on various onboard resources such as sleeping pods, food supply stations, oxygen refill units, medical bays, hygiene pods, and fitness centers.

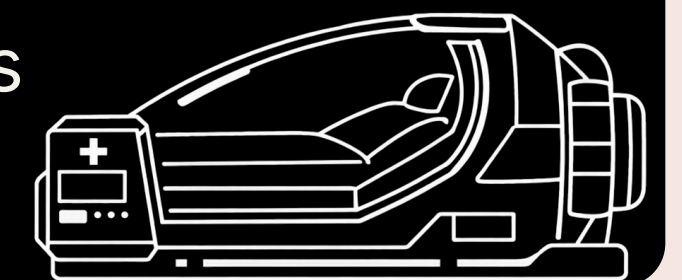
Three Crew Leads are responsible for managing passengers and the resources available inside the spaceship.

Passengers are assigned different membership levels, and access to resources depends on their level. The system must enforce access control rules while keeping track of resource usage.

Your task: Build a Passenger Resource Management System that allows Crew Leads to manage passengers/resources while allowing passengers to access permitted resources and track usage.

Critical Infrastructure

- Sleeping Pods
- Food Supply Stations
- Oxygen Refill Units
- Medical Bays
- Hygiene Pods & Fitness Centers



Core Objective: Passenger Levels

Rule: Higher levels inherit lower-level access

SILVER

Base tier access

Resources:

- Food Stations
- Sleeping Pods
- Basic Hygiene

Gold

Enhanced access

Resources:

- Private Cabins
- Adv. Medical Bay
- + All Silver

Platinum

Full ship access

Resources:

- Luxury O2 Pods
- VIP Rec Deck
- + All Gold & Silver

Administrative Control: The Crew Lead Role

System integrity is maintained by exactly three Crew Leads. These administrators hold exclusive permissions to manage the ship's complex ecosystem.

Only Crew Leads are authorized to perform administrative tasks, ensuring a secure and controlled environment for all passengers.

Exclusive Permissions

- Create and manage passenger profiles
- Upgrade or downgrade membership levels
- Provision and decommission ship resources
- Monitor real-time activity and usage reports



Passenger Experience: Seamless Interaction

Passengers interact with the PRMS through a streamlined interface designed for clarity and ease of use during the mission.

Resource Discovery

Instantly view all facilities available to your specific membership tier.



Active Usage

Seamlessly initiate the use of accessible resources with real-time validation.

Level 1: Basic Passenger & Resource Management

Crew Lead Enforcement

Strict system validation to ensure exactly three administrators.

Enforces safe limits to prevent unauthorized administrative expansion.

Resource Definition

Categorizing all ship resources by their minimum required membership level.

Establishing the base inventory for the mission.

Passenger Discovery

Enabling passengers to view a filtered list of accessible resources.

Ensuring transparency in available life-support services.

Level 2: Dynamic Access & Validation

The second phase introduces operational logic and state management, ensuring that resource usage is both secure and trackable.

This layer provides the Crew Leads with the tools to manage passenger tiers dynamically.

Permission Validation

Real-time checks performed before any resource is utilized to ensure strict adherence to membership rules.

Tier Management

Administrative tools for Crew Leads to upgrade or downgrade passenger levels based on mission requirements.

Audit Logging

Automatic recording of every resource interaction, creating a reliable audit trail for safety and reporting.

Level 3: Advanced Reporting & Insights

The final phase focuses on data-driven oversight, providing both passengers and Crew Leads with deep insights into resource consumption.

This ensures mission-critical resources are monitored and optimized for the long journey to Mars.

Personal History

Detailed logs for passengers to track their own resource consumption and well-being trends.



Aggregated Reports

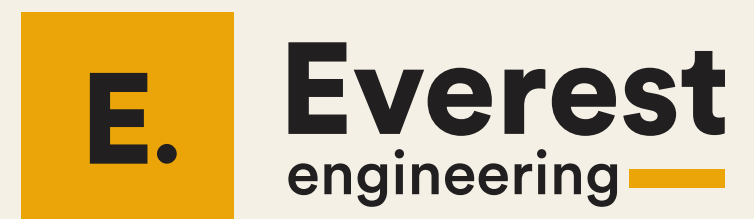
Summary views for Crew Leads, grouped by passenger level, to monitor ship-wide resource distribution.

Usage Analytics

Identifying high-demand resources like "Luxury Oxygen Pods" to prevent shortages.

SUMMARY: ENSURING MISSION SUCCESS

By implementing the Spaceship X26 PRMS, we ensure that every settler has the resources they need to thrive. The system provides the Crew Leads with the tools to manage the ship's complex ecosystem while offering passengers a transparent and reliable way to access life-support services.



THANK YOU

Safe Travels to the Red Planet

Spaceship X26 Mission Control

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